Quiz 2 formula sheet

ECET 250E · 6 November 2026

CH 7 First-order RC and RL circuits

Switching conditions

$$v_C(0^+) = v_C(0^-) = V_0$$

$$i_L(0^+) = i_L(0^-) = I_0$$

At DC steady state, open C and short L. Use the old circuit for initial values and the new circuit for final values.

$$i_C = C\,\frac{dv_C}{dt}, \qquad v_L = L\,\frac{di_L}{dt}$$

Current enters the positive voltage terminal.

Natural and step responses

$$\tau_{RC} = R_{\text{Th}}C, \qquad \tau_{RL} = L/R_{\text{Th}}$$

$$x(t) = x(\infty) + [x(0) - x(\infty)]e^{-t/\tau}$$

$$v(t) = V_0 e^{-t/\tau}, \qquad i(t) = I_0 e^{-t/\tau} \quad \text{(source-free)}$$

Find R_{th} looking into the element after switching with independent sources off. At a later switch use elapsed time.

$$w_L = \tfrac{1}{2}Li^2, \quad w_C = \tfrac{1}{2}Cv^2, \quad w_R = \int_{t_1}^{t_2} Ri^2\,dt$$

CH 8 Second-order RLC circuits

Roots and damping

$$\alpha_{\text{series}} = \frac{R}{2L}, \qquad \alpha_{\text{parallel}} = \frac{1}{2RC}$$

$$\omega_0 = \frac{1}{\sqrt{LC}}, \qquad s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2} \quad (\alpha < \omega_0)$$

Compare α with ω_0. Greater is overdamped, equal is critically damped, and smaller is underdamped.

Initial slopes for source-free circuits

Series: $\dfrac{di(0^+)}{dt} = -\dfrac{RI_0 + V_0}{L}$

Parallel: $\dfrac{dv(0^+)}{dt} = -\dfrac{V_0/R + I_0}{C}$

Series current enters C's + terminal. Parallel branch currents point from + to −. Reverse signs for reversed references.

Source-free series RLC

$$\alpha > \omega_0: \quad i(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

$$\alpha = \omega_0: \quad i(t) = (A_2 + A_1 t)e^{-\alpha t}$$

$$\alpha < \omega_0: \quad i(t) = e^{-\alpha t}(B_1 \cos\omega_d t + B_2 \sin\omega_d t)$$

Source-free parallel RLC

$$\alpha > \omega_0: \quad v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

$$\alpha = \omega_0: \quad v(t) = (A_1 + A_2 t)e^{-\alpha t}$$

$$\alpha < \omega_0: \quad v(t) = e^{-\alpha t}(A_1 \cos\omega_d t + A_2 \sin\omega_d t)$$

DC step response

Series: $v_C(t) = V_s + \text{transient}, \quad i = C\,dv_C/dt$

Parallel: $i_L(t) = I_s + \text{transient}, \quad v = L\,di_L/dt$

For a series voltage step or parallel current step, use the matching damping form from the parallel column above.

Solve for the constants

Set t = 0⁺ in the full response and its derivative. Match the stored value and initial slope.

Series step: $v_C(0^+) = V_0, \quad v_C'(0^+) = I_0/C$

Parallel step: $i_L(0^+) = I_0, \quad i_L'(0^+) = V_0/L$

CH 9 Sinusoids and phasors

Cosine reference

$$v(t) = V_m\cos(\omega t + \phi) \longleftrightarrow \mathbf{V} = V_m\angle\phi$$

$$\omega = 2\pi f, \quad T = 1/f, \quad \sin\theta = \cos(\theta - 90^\circ)$$

A negative amplitude adds 180°. At the same ω, wrap $\varphi_1 - \varphi_2$ to ±180°. A positive difference means signal 1 leads. For v(t), use radians for ωt and convert φ to radians.

Complex arithmetic

$$z = x + jy = r\angle\theta = r(\cos\theta + j\sin\theta)$$

$$r = \sqrt{x^2 + y^2}, \quad \theta = \operatorname{atan2}(y, x), \quad j^2 = -1$$

Add and subtract rectangular components. To multiply, multiply magnitudes and add angles. To divide, divide magnitudes and subtract angles. Add equal-frequency sinusoids as phasors.

Impedance and admittance

$$Z_R = R, \quad Z_L = j\omega L, \quad Z_C = \frac{1}{j\omega C}$$

$$Z = \mathbf{V}/\mathbf{I} = R + jX, \quad Y = 1/Z = G + jB$$

$$Z_{\text{series}} = \sum Z_k, \qquad Y_{\text{parallel}} = \sum Y_k$$

Voltage and current division

$$\mathbf{V}_x = \mathbf{V}_s\frac{Z_x}{\sum Z_k}, \qquad \mathbf{I}_1 = \mathbf{I}_T\frac{Z_2}{Z_1 + Z_2}$$

$$Z_1 \parallel Z_2 = \frac{Z_1 Z_2}{Z_1 + Z_2}$$

Voltage division uses series elements. Current division here uses two parallel branches. Return to time with the original ω.